

Non-calculator Part:

1. Find the inverse of matrix A given below.

$$\begin{array}{r} 27 \\ -15 \\ \hline 12 \end{array}$$

$$A = \begin{bmatrix} 4 & 3 & 5 \\ -5 & -3 & -9 \\ -1 & 0 & -3 \end{bmatrix}$$

$$\begin{bmatrix} 4 & 3 & 5 \\ -5 & -3 & -9 \\ -1 & 0 & -3 \end{bmatrix} \begin{array}{l} 15 \\ 36 \\ 27 \end{array}$$

$$(36 + 27) - (15 + 45)$$

$$63 - 70 = -7$$

$$-\frac{1}{7} \begin{bmatrix} 9 & 9 & -12 \\ -6 & -7 & 11 \\ -3 & -3 & 3 \end{bmatrix} = \begin{bmatrix} -9/7 & -9/7 & 12/7 \\ 6/7 & 1 & -11/7 \\ 3/7 & 3/7 & -3/7 \end{bmatrix}$$

$$\begin{bmatrix} 4 & 3 & 5 \\ -5 & -3 & -9 \\ -1 & 0 & -3 \end{bmatrix} \begin{array}{l} 15 \\ 36 \\ 27 \end{array}$$

$$\begin{bmatrix} 9 & -6 & -3 \\ 9 & -7 & -3 \\ -12 & 11 & 3 \end{bmatrix}^T = \begin{bmatrix} 9 & 9 & -12 \\ -6 & -7 & 11 \\ -3 & -3 & 3 \end{bmatrix}$$



$$-25 + 36 = 11$$

$$-27 + 15 = -12$$

$$A^{-1} = \begin{bmatrix} -9/7 & -9/7 & 12/7 \\ 6/7 & 1 & -11/7 \\ 3/7 & 3/7 & -3/7 \end{bmatrix}$$

2. Use Gauss-Jordan elimination to solve the following system of equations. You should indicate what specific row operation(s) you perform in each step and show all your work clearly. Hint: Your Analysis teachers are being nice and the answers are integers!

$$4x - 3y = 15$$

$$2x + 5y = 79$$

$$\left[\begin{array}{cc|c} 4 & -3 & 15 \\ 2 & 5 & 79 \end{array} \right] R_1 - 2R_2 \rightarrow \left[\begin{array}{cc|c} 4 & -3 & 15 \\ 0 & -13 & -143 \end{array} \right] 13R_1 - 3R_2$$

$$\left[\begin{array}{cc|c} 52 & 0 & 624 \\ 0 & -13 & -143 \end{array} \right] R_1 \cdot \frac{1}{52} \quad R_2 \cdot -\frac{1}{13}$$

$$\left[\begin{array}{cc|c} 1 & 0 & 12 \\ 0 & 1 & 11 \end{array} \right]$$

$$4-3$$

$$x = 12 \quad y = 11$$

$$\begin{array}{l} 4x - 3y = 15 \\ -4x + 10y = 158 \\ \hline 0 - 13y = 143 \\ y = 11 \end{array}$$

$$4x - 33 = 15$$

$$4x = 48$$

$$x = 12$$

$$\begin{array}{l} 4x - 3y = 15 \\ -2x + 5y = 79 \\ \hline \end{array}$$

$$\begin{array}{r} 13 \\ \times 11 \\ \hline 130 \\ 143 \\ \hline \end{array}$$

$$\begin{array}{r} 13 \\ \times 4 \\ \hline 52 \end{array}$$

$$\begin{array}{r} 52 \\ \times 12 \\ \hline 104 \\ 520 \\ \hline 624 \end{array}$$

$$\begin{array}{r} 143 \\ \times 3 \\ \hline 429 \end{array}$$

$$15 - 2(79) = 15 - 158 = -143$$

$$\begin{array}{r} 158 \\ -15 \\ \hline 143 \end{array}$$

$$\begin{array}{r} 79 \\ \times 2 \\ \hline 158 \end{array}$$

$$\begin{array}{r} 158 \\ -15 \\ \hline 143 \end{array}$$

12
12

3. Consider a Markov Chain represented by the transition matrix T below. How would you find the equilibrium vector of this Markov Chain algebraically? Write a 3×4 matrix that you would enter into a calculator such that its reduced row echelon form yields the equilibrium vector.

Do **NOT** solve the system.

$$T = \begin{bmatrix} 0.7 & 0.2 & 0.1 \\ 0.1 & 0.4 & 0.5 \\ 0 & 0.6 & 0.4 \end{bmatrix}$$

$$\begin{bmatrix} x & y & z \end{bmatrix} \begin{bmatrix} 0.7 & 0.2 & 0.1 \\ 0.1 & 0.4 & 0.5 \\ 0 & 0.6 & 0.4 \end{bmatrix} = \begin{bmatrix} x & y & z \end{bmatrix}$$

$$0.7x + 0.1y + 0 = x$$

$$0.2x + 0.4y + 0.6z = y$$

$$0.1x + 0.5y + 0.4z = z$$

$$-0.3x + 0.1y = 0$$

$$0.2x - 0.6y + 0.6z = 0$$

$$0.1x + 0.5y - 0.6z = 0$$

$$\left[\begin{array}{ccc|c} -0.3 & 0.1 & 0 & 0 \\ 0.2 & -0.6 & 0.6 & 0 \\ 1 & 1 & 1 & 1 \end{array} \right]$$



Calculator Part:

Write down the calculation that you entered into your calculator to solve each problem.

4. A research group is studying career movement among adults in a certain region. At the end of each year, a person can be in one of three states: **T** = works in the tech industry; **N** = works in a non-tech industry; **U** = unemployed.

Historical data suggests the following transition probabilities from one year to the next:

- A person working in **tech (T)** will
 - stay in tech with probability 0.7
 - move to a non-tech job with probability 0.2
 - become unemployed with probability 0.1
- A person working in **non-tech (N)** will
 - switch to tech with probability 0.15
 - stay in non-tech with probability 0.7
 - become unemployed with probability 0.15
- A person who is **unemployed (U)** will
 - find a tech job with probability 0.25
 - find a non-tech job with probability 0.5
 - remain unemployed with probability 0.25



$$A = \begin{matrix} & \begin{matrix} T & N & U \end{matrix} \\ \begin{matrix} T \\ N \\ U \end{matrix} & \begin{bmatrix} 0.7 & 0.2 & 0.1 \\ 0.15 & 0.7 & 0.15 \\ 0.25 & 0.5 & 0.25 \end{bmatrix} \end{matrix} \begin{matrix} = 1 \\ = 1 \\ = 1 \end{matrix}$$

- a) Complete the transition matrix *A* above right.
- b) What is the probability that a person who is working in tech will be unemployed in three years?

$$\begin{bmatrix} 0.7 & 0.2 & 0.1 \\ 0.15 & 0.7 & 0.15 \\ 0.25 & 0.5 & 0.25 \end{bmatrix}^3 = 0.13525 \quad \checkmark$$

- c) In the long run, what percentage of the population will be unemployed?

$$\begin{bmatrix} 0.7 & 0.2 & 0.1 \\ 0.15 & 0.7 & 0.15 \\ 0.25 & 0.5 & 0.25 \end{bmatrix}^{1000} = \begin{bmatrix} 0.365854 & 0.487805 & 0.146341 \end{bmatrix} \quad \boxed{0.146341} \quad \checkmark$$

- d) If the region has 50 thousand workers, how many of them will be working in the tech industry in the long run?

$$50 \cdot 0.365854 = 18.2927 \text{ thousand} \quad \checkmark$$

- e) After 4 years there are 17.06, 25.48 and 7.46 thousand workers respectively in the three states. How many thousands of workers were in each state after 1 year?

$$\begin{matrix} 17.06 \\ 25.48 \\ + 7.46 \\ \hline 50 \end{matrix}$$

$$[x \ y \ z] \begin{bmatrix} 0.7 & 0.2 & 0.1 \\ 0.15 & 0.7 & 0.15 \\ 0.25 & 0.5 & 0.25 \end{bmatrix}^4 = [17.06 \ 25.48 \ 7.46]$$

$$\begin{matrix} X=4 \\ Y=36 \\ Z=10 \end{matrix}$$

$$\begin{matrix} 0.42x + 0.33y + 0.35z = 17.06 \\ 0.44x + 0.52y + 0.60z = 25.48 \\ 0.14x + 0.15y + 0.15z = 7.46 \end{matrix} \rightarrow [4 \ 36 \ 10] \begin{bmatrix} 0.7 & 0.2 & 0.1 \\ 0.15 & 0.7 & 0.15 \\ 0.25 & 0.5 & 0.25 \end{bmatrix}^1 = [10.7 \ 31 \ 8.3]$$

$$\boxed{10.7, 31, 8.3}$$

you should not use rounded